

Samartha Shrestha

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PROFILE

Computer Science undergraduate with hands-on experience building software and AI/ML applications using Python, JavaScript, React Native, and FastAPI. Experienced in developing practical solutions, working with APIs and machine learning workflows, and collaborating on real-world projects. Seeking a full-time software engineering opportunity in backend or full-stack development.

EDUCATION

Symbiosis Institute of Technology

Aug 2023 – Jul 2027

B.Tech in Computer Science

Pune, India

Relevant Coursework: Data Structures, Object-Oriented Programming, Artificial Intelligence, Database Concepts, Cybersecurity Basics, Internet Technologies

PROJECTS

LUMEN

AI/ML, YOLO, FastAPI, React Native

AI-powered civic infrastructure system that detects road damage from user-uploaded images, helping identify and report road-related issues more efficiently.

- Developed LUMEN as an AI/ML-based road-damage detection system using a trained YOLO model to identify road issues from uploaded images.
- Integrated the machine learning detection pipeline with a FastAPI backend and React Native client to create an end-to-end image-based detection workflow.
- Applied iterative problem solving and development practices to build a practical technology solution for identifying and reporting civic infrastructure issues.

Finoryx

React Native, JavaScript, GitHub

- Developed a mobile application focused on delivering a clean and intuitive user experience.
- Translated application requirements into reusable frontend components and functional workflows using JavaScript and React Native.
- Used GitHub for source control and collaborative development, maintaining structured project workflows.

Cypher-Flux

React, Vite, Tailwind CSS, Python, Flask, Socket.io

- Developed a real-time Security Operations Center (SOC) dashboard for threat monitoring, anomaly detection, alert management, and security response.
- Built a decoupled React frontend and Python Flask backend with Socket.io for real-time security alerts, dynamic threat metrics, and live network monitoring.

Advanced LMS – NGO Project

Web Technologies

- Developed a Learning Management System designed for government school students, focusing on practical user requirements.
- Collaborated with Seer Bharat Foundation to understand real-world requirements and translate them into application functionality.

SKILLS

AI & Technology: Artificial Intelligence, AI/ML, AI-assisted workflows, Data Interpretation

Technical: Python, JavaScript, React Native, HTML, CSS, C/C++

Core: Data Structures, OOP, Database Concepts, Cybersecurity Basics

Tools: FastAPI, Git, GitHub, VS Code

HONORS & ACHIEVEMENTS

- **AWS Certified Cloud Practitioner** – Amazon Web Services (AWS)
- **Coursera** – Cloud Computing
- **IBM Machine Learning Professional Certificate** – IBM
- **Hacker's Gambit (CTF)** – Participated in Capture The Flag cybersecurity challenges, Oct 2025.
- **Huntress CTF** – Participated in cybersecurity challenge-based problem solving, Oct 2025.

SOFT SKILLS

Communication

Teamwork

Problem Solving

Time Management

Adaptability

Critical Thinking